

OUTPUT-

1. Adam is working in an IT company. He has been given a task to reduce the load of a system by kitling some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?

- **Kill Processes by name**

```
who@Waghmare08:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1      0  0 14:49 ?        00:00:00 /sbin/init
root           2      1  0 14:49 ?        00:00:00 /init
root           8      2  0 14:49 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          62      1  0 14:49 ?        00:00:00 /lib/systemd/systemd-journald
root          88      1  0 14:49 ?        00:00:00 /lib/systemd/systemd-udev
systemd+      91      1  0 14:49 ?        00:00:00 /lib/systemd/systemd-resolved
systemd+      92      1  0 14:49 ?        00:00:00 /lib/systemd/systemd-timesyncd
root         193      1  0 14:49 ?        00:00:00 /usr/sbin/cron -f -P
message+     196      1  0 14:49 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root         201      1  0 14:49 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
syslog       202      1  0 14:49 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         205      1  0 14:49 ?        00:00:00 /lib/systemd/systemd-logind
root         227      1  0 14:49 hvc0    00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud console 115200,38400,9600 vt220
root         229      1  0 14:49 tty1    00:00:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root         230      1  0 14:49 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         294      2  0 14:49 ?        00:00:00 /init
root         295     294  0 14:49 ?        00:00:00 /init
who          296     295  0 14:49 pts/0    00:00:00 -bash
root         297      2  0 14:49 pts/1    00:00:00 /bin/login -f
who          340      1  0 14:49 ?        00:00:00 /lib/systemd/systemd --user
who          341     340  0 14:49 ?        00:00:00 (sd-pam)
who          357     297  0 14:49 pts/1    00:00:00 -bash
who          447     296  0 14:57 pts/0    00:00:00 ps -ef
who@Waghmare08:~$ pkill who
who@Waghmare08:~$ pkill root
who@Waghmare08:~$ pkill payal
who@Waghmare08:~$ ps -ef|payal
payal: command not found
who@Waghmare08:~$ ps -ef
```

- **Kill a process based on the process name**

```

who@Waghmare08:~$ ps -ef
root      229      1  0 14:49 tty1      00:00:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root      230      1  0 14:49 ?                00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root      294      2  0 14:49 ?                00:00:00 /init
root      295      294  0 14:49 ?                00:00:00 /init
who        296      295  0 14:49 pts/0          00:00:00 -bash
root      297      2  0 14:49 pts/1          00:00:00 /bin/login -f
who        340      1  0 14:49 ?                00:00:00 /lib/systemd/systemd --user
who        341      340  0 14:49 ?                00:00:00 (sd-pam)
who        357      297  0 14:49 pts/1          00:00:00 -bash
who        447      296  0 14:57 pts/0          00:00:00 ps -ef
who@Waghmare08:~$ kill who
who@Waghmare08:~$ kill root
who@Waghmare08:~$ kill payal
who@Waghmare08:~$ ps -ef|payal
payal: command not found
who@Waghmare08:~$ ps -ef
UID        PID     PPID  C STIME TTY          TIME CMD
root         1         0  0 14:49 ?                00:00:00 /sbin/init
root         2         1  0 14:49 ?                00:00:00 /init
root         8         2  0 14:49 ?                00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        62         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-journald
root        88         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-udevd
systemd+    91         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-resolved
systemd+    92         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-timesyncd
root       193         1  0 14:49 ?                00:00:00 /usr/sbin/cron -f -P
message+   196         1  0 14:49 ?                00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root       201         1  0 14:49 ?                00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
syslog     202         1  0 14:49 ?                00:00:00 /usr/sbin/rsyslogd -n -iNONE
root       205         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-logind
root       227         1  0 14:49 hvc0          00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud console 115200,38400,9600 vt220
root       229         1  0 14:49 tty1          00:00:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root       230         1  0 14:49 ?                00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root       294         2  0 14:49 ?                00:00:00 /init
root       295      294  0 14:49 ?                00:00:00 /init
who        296      295  0 14:49 pts/0          00:00:00 -bash
root       297         2  0 14:49 pts/1          00:00:00 /bin/login -f
who        340         1  0 14:49 ?                00:00:00 /lib/systemd/systemd --user
who        341      340  0 14:49 ?                00:00:00 (sd-pam)
who        357      297  0 14:49 pts/1          00:00:00 -bash
who        465      296  0 14:58 pts/0          00:00:00 ps -ef
who@Waghmare08:~$ sleep 300 & [1] 468
[1] 468
[1]: command not found
who@Waghmare08:~$ sleep 300 & [1] 468
[2] 482
[1]: command not found
who@Waghmare08:~$ |

```

- Kill a single process at a time with the given process ID

```

who@Waghmare08:~$ ps -ef
UID        PID     PPID  C STIME TTY          TIME CMD
root         1         0  0 14:49 ?                00:00:00 /sbin/init
root         2         1  0 14:49 ?                00:00:00 /init
root         8         2  0 14:49 ?                00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        62         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-journald
root        88         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-udevd
systemd+    91         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-resolved
systemd+    92         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-timesyncd
root       193         1  0 14:49 ?                00:00:00 /usr/sbin/cron -f -P
message+   196         1  0 14:49 ?                00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root       201         1  0 14:49 ?                00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
syslog     202         1  0 14:49 ?                00:00:00 /usr/sbin/rsyslogd -n -iNONE
root       205         1  0 14:49 ?                00:00:00 /lib/systemd/systemd-logind
root       227         1  0 14:49 hvc0          00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud console 115200,38400,9600 vt220
root       229         1  0 14:49 tty1          00:00:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root       230         1  0 14:49 ?                00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root       294         2  0 14:49 ?                00:00:00 /init
root       295      294  0 14:49 ?                00:00:00 /init
who        296      295  0 14:49 pts/0          00:00:00 -bash
root       297         2  0 14:49 pts/1          00:00:00 /bin/login -f
who        340         1  0 14:49 ?                00:00:00 /lib/systemd/systemd --user
who        341      340  0 14:49 ?                00:00:00 (sd-pam)
who        357      297  0 14:49 pts/1          00:00:00 -bash
who        468      296  0 14:59 pts/0          00:00:00 sleep 300
who        482      296  0 15:00 pts/0          00:00:00 sleep 300
who        495      296  0 15:02 pts/0          00:00:00 ps -ef
who@Waghmare08:~$ ps -ef|grep sleep
who        468      296  0 14:59 pts/0          00:00:00 sleep 300
who        482      296  0 15:00 pts/0          00:00:00 sleep 300
who        499      296  0 15:02 pts/0          00:00:00 grep --color=auto sleep
who@Waghmare08:~$ killall sleep
[1]- Terminated                  sleep 300
[2]+ Terminated                  sleep 300
who@Waghmare08:~$ |

```

2. Write a program for process creation using C

- **Orphan Process**

```
who@Waghmare08:~$ nano orphan.c
who@Waghmare08:~$ gcc orphan.c -o orphan
who@Waghmare08:~$ ./orphan
Parent process exiting
who@Waghmare08:~$ cChild process running
PID = 1492
PPID = 295
|
```

- **Zombie Process**

```
who@Waghmare08:~$ nano zombie.c
who@Waghmare08:~$ gcc zombie.c -o zombie
who@Waghmare08:~$ ./zombie
Child process exiting
Parent process running
who@Waghmare08:~$ |
```

3. Create the process using fork 0 system cgl' ‘

- **Child Process creation**
- **Parent Process creation**

```
who@Waghmare08:~$ nano fork.c
who@Waghmare08:~$ gcc fork.c -o fork
who@Waghmare08:~$ ./fork
Parent Process
PID = 1551
Child PID = 1552
Child Process
PID = 1552
PPID = 1551
who@Waghmare08:~$ |
```

- **PPID and PID**

```
who@Waghmare08:~$ sleep 300 & [1] 2754
[1] 1556
[1]: command not found
who@Waghmare08:~$ ps -ef|grep sleep
who    1556    296  0 15:41 pts/0    00:00:00 sleep 300
who    1569    296  0 15:41 pts/0    00:00:00 grep --color=auto sleep
who@Waghmare08:~$ kill 2754
-bash: kill: (2754) - No such process
who@Waghmare08:~$ killall sleep
[1]+  Terminated          sleep 300
who@Waghmare08:~$ |
```